

The Ornstein–Uhlenbeck generator domain need not lie in the drift domain

Candidate full answer to the domain question in arXiv:math/0509309

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Status: candidate full counterexample, likely valid. There is an Ornstein–Uhlenbeck semigroup P on a real Hilbert space for which

$$\mathcal{D}(L_{P^\circ}) \not\subseteq \mathcal{D}(L_{R^\circ}).$$

In fact, the witness is a bounded uniformly continuous function fixed by P , and therefore has P° -generator zero, while its pointwise drift derivatives are unbounded.

1 The source question

Let S be a C_0 -semigroup on a real Banach space E , let P be the associated Ornstein–Uhlenbeck semigroup, and let

$$R(t)f(x) = f(S(t)x)$$

be the drift semigroup. Van Neerven and Priola prove that the maximal closed subspaces of $\text{BUC}(E)$ on which P and R are strongly continuous coincide; they call this common space $\text{BUC}^\circ(E)$. They then ask whether the generators of the restricted C_0 -semigroups always satisfy

$$\mathcal{D}(L_{P^\circ}) \subseteq \mathcal{D}(L_{R^\circ}) \tag{Q}$$

[1, Section 2]. We give a negative answer.

2 A finite-dimensional building block

Let $d \geq 3$, $q > 0$, and $\sigma^2 = q/2$. In $\mathbb{R}^d = \mathbb{R} \oplus \mathbb{R}^{d-1}$ consider the homogeneous cone

$$C_d = \{(z_1, z') : z_1 \geq \|z'\|\}$$

and define

$$F_{d,q}(u) = \int_{\mathbb{R}^d} \mathbf{1}_{C_d}(u + y) d\mathcal{N}(0, \sigma^2 I_d)(y). \tag{1}$$

Thus $0 \leq F_{d,q} \leq 1$. Gaussian convolution makes $F_{d,q}$ smooth and bounded uniformly continuous.

Lemma 1 (Harmonic cone block). *Consider the outward Ornstein–Uhlenbeck semigroup on $\mathbb{R}^d$ with $S(t) = e^t I_d$ and noise covariance qI_d . Then*

$$P(t)F_{d,q} = F_{d,q} \quad (t \geq 0).$$

Moreover, with $m_d = \mathbb{E}\|Z'\|$ for $Z' \sim \mathcal{N}(0, I_{d-1})$, one has at $u_d = \sigma m_d e_1$

$$u_d \cdot \nabla F_{d,q}(u_d) \geq c_0 \sqrt{d-2}, \quad c_0 = \frac{3}{4}(2\pi)^{-1/2} e^{-2}. \quad (2)$$

The constant is independent of d and q .

Proof intuition. The extra Gaussian in (1) is chosen to be the covariance accumulated by running the outward OU flow backwards from infinity. After an OU step, the step noise plus this extra Gaussian is simply an e^t -dilate of the original Gaussian; the cone indicator is unchanged by that dilation. On a d -dimensional block, crossing the cone boundary occurs in a radial window of fixed Gaussian width at radius comparable to $\sqrt{d}$. Multiplication by the radial drift therefore produces a derivative of order $\sqrt{d}$.

Proof. The OU noise at time t has covariance

$$Q_t = \int_0^t e^{2s} q I_d ds = \frac{e^{2t} - 1}{2} q I_d.$$

Let $Y_t \sim \mathcal{N}(0, Q_t)$ and $Y \sim \mathcal{N}(0, qI_d/2)$ be independent. Their sum has the same law as $e^t Y$. Since $\mathbf{1}_{C_d}(e^t z) = \mathbf{1}_{C_d}(z)$ for $t \in \mathbb{R}$,

$$\begin{aligned} P(t)F_{d,q}(u) &= \mathbb{E} \mathbf{1}_{C_d}(e^t u + Y_t + Y) \\ &= \mathbb{E} \mathbf{1}_{C_d}(e^t(u + Y)) = F_{d,q}(u). \end{aligned}$$

For the derivative estimate, write $Z = (Z_1, Z') \sim \mathcal{N}(0, I_d)$, $T = \|Z'\|$, and let Φ, ϕ be the standard normal distribution function and density. Along the first coordinate axis,

$$F_{d,q}(r e_1) = \mathbb{P}\{r + \sigma Z_1 \geq \sigma T\} = \mathbb{E}\Phi(r/\sigma - T).$$

Differentiation under the expectation gives

$$\frac{d}{dr} F_{d,q}(r e_1) = \sigma^{-1} \mathbb{E}\phi(r/\sigma - T).$$

At $r = \sigma m_d$ it follows that

$$r \frac{d}{dr} F_{d,q}(r e_1) = m_d \mathbb{E}\phi(m_d - T). \quad (3)$$

The Gaussian Poincaré inequality applied to the 1-Lipschitz map $z' \mapsto \|z'\|$ gives $\text{Var}(T) \leq 1$. Hence

$$m_d^2 = \mathbb{E}T^2 - \text{Var}(T) \geq (d-1) - 1 = d-2.$$

Chebyshev's inequality gives $\mathbb{P}(|T - m_d| \leq 2) \geq 3/4$, and on this event $\phi(m_d - T) \geq \phi(2)$. Substitution in (3) proves (2). $\square$

3 The Hilbert-space counterexample

Let

$$E = \left(\bigoplus_{n \geq 1} \mathbb{R}^{d_n} \right)_{\ell^2}, \quad d_n = 2^{4n}, \quad q_n = 2^{-6n}.$$

Let $A = I_E$, so $S(t) = e^t I_E$, and let the positive covariance operator Q act as $q_n I_{d_n}$ on the n -th block. It is trace class because

$$\text{Tr } Q = \sum_{n \geq 1} d_n q_n = \sum_{n \geq 1} 2^{-2n} < \infty.$$

Taking $B = Q^{1/2}$ gives legitimate centred Gaussian Radon measures with covariance

$$Q_t = \frac{e^{2t} - 1}{2} Q$$

for every $t \geq 0$.

Let $\pi_n : E \rightarrow \mathbb{R}^{d_n}$ be the coordinate projection, put $F_n = F_{d_n, q_n}$, and define

$$f(x) = \sum_{n=1}^{\infty} 2^{-n} F_n(\pi_n x). \quad (4)$$

The series converges uniformly because $0 \leq F_n \leq 1$. Each summand is bounded uniformly continuous, so $f \in \text{BUC}(E)$.

Theorem 2. *For the preceding Ornstein–Uhlenbeck and drift semigroups,*

$$f \in \text{D}(L_{P^\circ}), \quad L_{P^\circ} f = 0, \quad \text{but} \quad f \notin \text{D}(L_{R^\circ}).$$

Consequently the inclusion (Q) fails in general.

Proof. The n -th coordinate of the OU noise has exactly the finite-dimensional law used in Lemma 1. Hence every summand in (4) is fixed by $P(t)$. Uniform convergence and contractivity allow the sum to pass through $P(t)$, giving

$$P(t)f = f \quad (t \geq 0).$$

Thus f lies in the strong-continuity space $\text{BUC}^\circ(E)$ and in $\text{D}(L_{P^\circ})$, with generator zero.

Suppose, toward a contradiction, that $f \in \text{D}(L_{R^\circ})$, and set $h = L_{R^\circ} f \in \text{BUC}^\circ(E)$. For a point x supported only in the n -th block, say $\pi_n x = u$ and $\pi_k x = 0$ for $k \neq n$, all summands but the n -th are constant along $t \mapsto e^t x$. The pointwise generator limit therefore gives

$$h(x) = 2^{-n} u \cdot \nabla F_n(u).$$

Choose $u = \sqrt{q_n/2} m_{d_n} e_1$. Lemma 1 yields

$$\|h\|_\infty \geq 2^{-n} c_0 \sqrt{d_n - 2} = 2^{-n} c_0 \sqrt{2^{4n} - 2} \rightarrow \infty.$$

This contradicts the boundedness of h . Therefore $f \notin \text{D}(L_{R^\circ})$. □

4 Audit, scope, and novelty

The construction uses a bounded generator $A = I$, a positive trace-class covariance, and genuine Gaussian Radon transition measures; no cylindrical or non-Radon noise is hidden in the example. The witness is real-valued, bounded, and uniformly continuous. Membership in the common space $\text{BUC}^\circ(E)$ follows directly from $P(t)f = f$ and the source's proved equality of the P - and R -strong-continuity spaces. Failure of the drift domain is stronger than failure of a formal differentiability condition: the pointwise drift derivatives forced by any generator limit are unbounded.

The result answers only the word “always” in the source question. It does not preclude the inclusion under finite-dimensional ellipticity, analyticity, or other regularity hypotheses. Exact-pharse, title-citation, generator-domain, and later-survey searches through 11 August 2026 did not locate a published answer to this specific question. The few later papers found citing the source use its norm-discontinuity or spectral results and do not state this counterexample. Novelty confidence is moderate; independent expert review is requested.

References

- [1] J. M. A. M. van Neerven and E. Priola, *Norm discontinuity and spectral properties of Ornstein–Uhlenbeck semigroups*, Journal of Evolution Equations 5 (2005), 557–576; arXiv:math/0509309, <https://arxiv.org/abs/math/0509309>.